Formulations of coloring problems

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Learning objectives

At the end of this chapter you will be able to:

- write the ILP model of the Vertex Coloring Problem and explain the role of the activation variables y_c;
- write the two LP relaxations and their duals, and compute their optimal solutions;
- prove that the optimal value of the LP relaxations is 2 for every graph with an edge;
- explain why this bound is very weak, and why the model is not enough to color a graph.

Prerequisites

- From the module *Linear Programming*, the chapter «Duality theory»: the dual of an LP problem, weak and strong duality
- From the module *Integer linear programming*, the chapters «Integer linear programming» and «Primal and dual bounds»: ILP problems, LP relaxations and bounds
- From this module, the chapter «Coloring problems»: the problem, its assumptions and the notation on graphs

Overview of the family

Introduction: Overview of the family

- In this section we see the common structure of the **ILP models** of the coloring problems
- All of them use an upper bound k on the number of colors, assignment variables and activation variables
- Recognizing this structure helps to read every model of the family, and to compare it with the models of the packing problems

The ingredients of the coloring models

- The ILP models of the coloring problems use an upper bound k on the number of colors and the set $\mathscr{C} = \{1, 2, \ldots, k\}$ of the colors that may be used
- The binary **assignment variables** x_{vc} give a color $c \in \mathscr{C}$ to every vertex $v \in V$, and the binary **activation variables** y_c say whether color c is used
- Each edge and each color give one constraint, which forbids the same color on the two endpoints of the edge, and the objective function minimizes the number $\sum_{c \in \mathscr{C}} y_c$ of colors used

Coloring models are assignment models

The coloring models are **assignment models**, as the packing models: a vertex is assigned to a color as an item is assigned to a bin, and the edge constraints replace the capacity constraints.

Coloring models	**Packing models**
a vertex	an item
a color	a bin
the edge constraints	the capacity constraints

The members of the family

- The table lists the members of the family, with the name of their ILP model and the section that writes it

Problem	Model	Variables	Section
VCP	$\mathrm{ILP}_{\mathrm{VCP}}$	binary x_{vc} and y_c	An ILP model for the Vertex Coloring Problem

Summary

The ingredients:

- An upper bound k on the number of colors and the set $\mathscr{C} = \{1, 2, \ldots, k\}$ of the colors
- Assignment variables x_{vc} and activation variables y_c, all binary
- One constraint for each edge and each color; the objective function counts the colors used

Assignment models:

- A vertex is assigned to a color as an item is assigned to a bin
- The edge constraints replace the capacity constraints of the packing models

An ILP model for the Vertex Coloring Problem

Introduction: An ILP model for the VCP

- The Vertex Coloring Problem and its assumptions are defined in the chapter «Coloring problems»
- In this section we introduce the variables, write the $\mathrm{ILP}_{\mathrm{VCP}}$ model and describe its objective function and its constraints
- We explain the link between the assignment variables and the activation variables
- We write the model for a small graph, a triangle: it is the instance used in the whole chapter

The variables of the model (1/2)

- Given a graph $G = (V, E)$, for a vertex $v \in V$ we denote by $\delta(v)$ the set of the edges incident to v

- Given an upper bound $k \in \mathbb{Z}_{\geq 2}$ on the number of colors used in any optimal VCP solution, we introduce the following $k \underbrace{|V|}_{=n}$ binary variables:

$$x_{vc} = \begin{cases} 1 & \text{if vertex } v \text{ takes color } c, \\ 0 & \text{otherwise} \end{cases}$$

$$\forall v \in \underbrace{\{1, 2, \ldots, n\}}_{=V}, \ \forall c \in \underbrace{\{1, 2, \ldots, k\}}_{=\mathscr{C}},$$

The variables of the model (2/2)

- and the following k binary variables:

$$y_c = \begin{cases} 1 & \text{if color } c \text{ is used,} \\ 0 & \text{otherwise} \end{cases} \qquad \forall c \in \underbrace{\{1, 2, \ldots, k\}}_{=\mathscr{C}}.$$

- The number n of vertices is a valid upper bound on the minimum number of colors needed to color all the vertices since, in the worst possible case, each vertex takes a different color

- Also the number of colors used in any feasible VCP solution is a valid upper bound

The $\mathrm{ILP}_{\mathrm{VCP}}$ model

Using these variables, an Integer Linear Programming (ILP) model for the Vertex Coloring Problem (VCP) reads as follows:

$$\min \quad \sum_{c \in \mathscr{C}} y_c$$

$$\text{subject to} \quad \sum_{c \in \mathscr{C}} x_{vc} = 1, \qquad \forall v \in V,$$

$$-x_{uc} - x_{vc} + y_c \geq 0, \qquad \forall c \in \mathscr{C},\ \forall \{u, v\} \in E,$$

$$x_{vc} \in \{0, 1\}, \qquad \forall v \in V,\ \forall c \in \mathscr{C},$$

$$y_c \in \{0, 1\}, \qquad \forall c \in \mathscr{C}.$$

This ILP model is referred to as the $\mathrm{ILP}_{\mathrm{VCP}}$ model.

The objective function and the constraints

Description of the objective function and constraints of the $\mathrm{ILP}_{\mathrm{VCP}}$ model:

- the linear objective function $\min \sum_{c \in \mathscr{C}} y_c$ minimizes the total number of used colors;
- the constraints $\sum_{c \in \mathscr{C}} x_{vc} = 1$ ensure that each vertex takes exactly one color (n linear constraints);
- the constraints $-x_{uc} - x_{vc} + y_c \geq 0$ ensure that, for each color and each edge, the two adjacent vertices cannot both take that color, and that an unused color is taken by no vertex ($k \underbrace{|E|}_{=m}$ linear constraints);
- finally, the constraints $x_{vc} \in \{0, 1\}$ and $y_c \in \{0, 1\}$ define the variables of the ILP model.

The edge constraints tie together
the assignment variables x_{vc} and the activation variables y_c.

Let us see this link in detail.

The link between the variables (1/2)

- **If color c is assigned to a vertex, then it is used.** The logical implication is

$$(x_{1c} \text{ OR } x_{2c} \text{ OR } \ldots \text{ OR } x_{nc}) \quad \Longrightarrow \quad y_c.$$

  Every vertex has an incident edge $\{u, v\}$, and the constraint $x_{uc} + x_{vc} \leq y_c$ forces $y_c = 1$ as soon as $x_{vc} = 1$.

- **If color c is not used, then no vertex takes it.** This is the contrapositive of the previous implication. For $y_c = 0$ the edge constraint gives $x_{uc} = x_{vc} = 0$ for every edge $\{u, v\} \in E$.

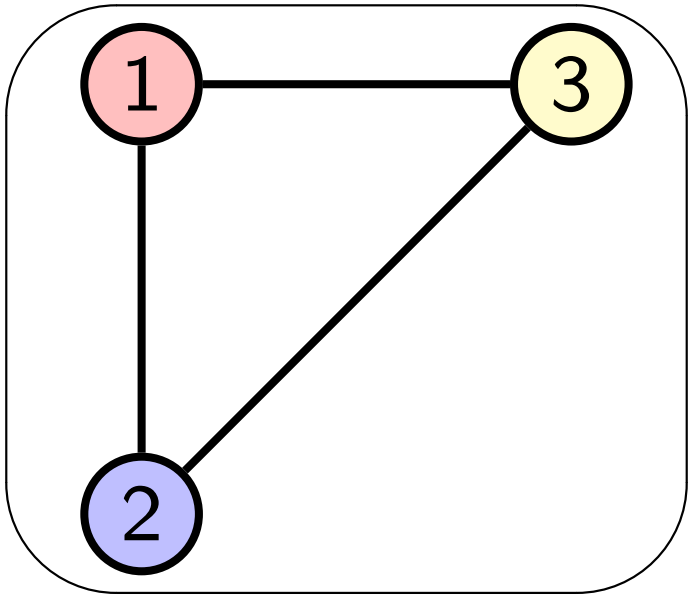
The link between the variables (2/2)

- **If color c is used, then some vertex takes it, but only in an optimal solution.**
- The constraints do not impose it
- However, if $x_{vc} = 0$ for all v and $y_c = 1$, setting $y_c = 0$ keeps all the constraints satisfied and decreases the objective value by one
- Hence, in every optimal solution, unused colors are not counted

We now write the $\mathrm{ILP}_{\mathrm{VCP}}$ model
for a small instance: a triangle.

Example: the $\mathrm{ILP}_{\mathrm{VCP}}$ model (1/4)

We consider the following graph $G = (V, E)$ with 3 vertices ($n = 3$) and 3 edges ($m = 3$):



The vertex set V and the edge set E are:

$$V = \{\ 1,\ 2,\ 3\ \}, \qquad E = \{\ \{1,2\},\ \{1,3\},\ \{2,3\}\ \}.$$

We refer to this VCP instance as the **triangle** example.

Example: the $\mathrm{ILP}_{\mathrm{VCP}}$ model (2/4)

The $\mathrm{ILP}_{\mathrm{VCP}}$ model with $k = 3$ for this instance is:

$$
\begin{array}{rcccccccccccccc}
\min & & & & & & & & & & y_1 & + y_2 & + y_3 & & \\
\text{subject to} & x_{11} & + x_{12} & + x_{13} & & & & & & & & & & & = 1 \\
 & & & & x_{21} & + x_{22} & + x_{23} & & & & & & & & = 1 \\
 & & & & & & & x_{31} & + x_{32} & + x_{33} & & & & & = 1 \\
 & -x_{11} & & & -x_{21} & & & & & & + y_1 & & & & \geq 0 \\
 & -x_{11} & & & & & & -x_{31} & & & + y_1 & & & & \geq 0 \\
 & & & & -x_{21} & & & -x_{31} & & & + y_1 & & & & \geq 0 \\
 & & -x_{12} & & & -x_{22} & & & & & & + y_2 & & & \geq 0 \\
 & & -x_{12} & & & & & & -x_{32} & & & + y_2 & & & \geq 0 \\
 & & & & & -x_{22} & & & -x_{32} & & & + y_2 & & & \geq 0 \\
 & & & -x_{13} & & & -x_{23} & & & & & & + y_3 & & \geq 0 \\
 & & & -x_{13} & & & & & & -x_{33} & & & + y_3 & & \geq 0 \\
 & & & & & & -x_{23} & & & -x_{33} & & & + y_3 & & \geq 0 \\
 & x_{11}, & x_{12}, & x_{13}, & x_{21}, & x_{22}, & x_{23}, & x_{31}, & x_{32}, & x_{33}, & y_1, & y_2, & y_3 & & \in \{0,1\}
\end{array}
$$

Example: the $\mathrm{ILP}_{\mathrm{VCP}}$ model (3/4)

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP}_{\mathrm{VCP}})$ is:

$$\tilde{x}_{11} = 1, \quad \tilde{x}_{12} = \tilde{x}_{13} = \tilde{x}_{21} = 0,$$

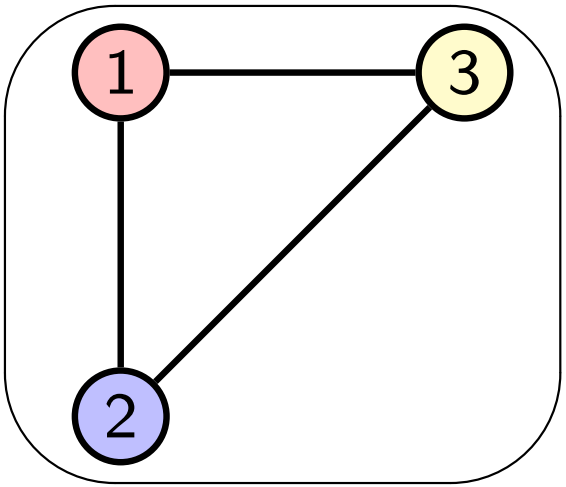
$$\tilde{x}_{22} = 1, \quad \tilde{x}_{23} = \tilde{x}_{31} = \tilde{x}_{32} = 0,$$

$$\tilde{x}_{33} = 1,$$

$$\tilde{y}_1 = \tilde{y}_2 = \tilde{y}_3 = 1$$

The optimal value $z(\mathrm{ILP}_{\mathrm{VCP}})$ is 3.

Example: the $\mathrm{ILP}_{\mathrm{VCP}}$ model (4/4)



- The associated optimal solution of the VCP is given by the color classes

$$\tilde{Q}_1 = \{1\}, \quad \tilde{Q}_2 = \{2\}, \quad \tilde{Q}_3 = \{3\}$$

- The optimal partition is unique, but the 3! permutations of the colors give 6 different optimal solutions

Summary

The $\mathrm{ILP}_{\mathrm{VCP}}$ model:

- $k\,n$ binary variables x_{vc} (vertex v takes color c) and k binary variables y_c (color c is used)
- n constraints give one color to each vertex; $k\,m$ edge constraints forbid equal colors on adjacent vertices

The link between the variables:

- A color taken by a vertex is used; an unused color is taken by no vertex
- A color that no vertex takes is switched off by the objective function, in every optimal solution

The triangle: $z(\mathrm{ILP}_{\mathrm{VCP}}) = 3$, with 6 optimal solutions that differ by a permutation of the colors

A first Linear Programming relaxation

Introduction: A first Linear Programming relaxation

- In this section we relax the binary variables of the $\mathrm{ILP}_{\mathrm{VCP}}$ model into non-negative variables: we obtain the $\mathrm{LP\text{-}0}_{\mathrm{VCP}}$ model
- Its optimal value is a lower bound on $z(\mathrm{ILP}_{\mathrm{VCP}})$: we compute it on the triangle
- We write the dual model $D(\mathrm{LP\text{-}0}_{\mathrm{VCP}})$ and solve it on the triangle
- Finally we prove, in two ways, that the optimal value is 2 for every graph with an edge

The LP-0_{VCP} model

To obtain a Linear Programming (LP) relaxation of the $\mathrm{ILP}_{\mathrm{VCP}}$ model, we relax the constraints $x_{vc} \in \{0, 1\}$ and $y_c \in \{0, 1\}$ with the following ones:

$$x_{vc} \geq 0, \quad \forall v \in V, \ \forall c \in \mathscr{C},$$

$$y_c \geq 0, \qquad \forall c \in \mathscr{C}.$$

This LP model is referred to as the LP-0_{VCP} model.

Example: the LP-0_{VCP} model (1/2)

The LP-0_{VCP} model for the triangle example is:

$$
\begin{array}{rccccccccccccc}
\min & & & & & & & & & & y_1 & + y_2 & + y_3 & \\
\text{subject to} & x_{11} & + x_{12} & + x_{13} & & & & & & & & & & = 1 \\
 & & & & x_{21} & + x_{22} & + x_{23} & & & & & & & = 1 \\
 & & & & & & & x_{31} & + x_{32} & + x_{33} & & & & = 1 \\
 & -x_{11} & & & -x_{21} & & & & & & + y_1 & & & \geq 0 \\
 & -x_{11} & & & & & & -x_{31} & & & + y_1 & & & \geq 0 \\
 & & & & -x_{21} & & & -x_{31} & & & + y_1 & & & \geq 0 \\
 & & -x_{12} & & & -x_{22} & & & & & & + y_2 & & \geq 0 \\
 & & -x_{12} & & & & & & -x_{32} & & & + y_2 & & \geq 0 \\
 & & & & & -x_{22} & & & -x_{32} & & & + y_2 & & \geq 0 \\
 & & & -x_{13} & & & -x_{23} & & & & & & + y_3 & \geq 0 \\
 & & & -x_{13} & & & & & & -x_{33} & & & + y_3 & \geq 0 \\
 & & & & & & -x_{23} & & & -x_{33} & & & + y_3 & \geq 0 \\
 & x_{11}, & x_{12}, & x_{13}, & x_{21}, & x_{22}, & x_{23}, & x_{31}, & x_{32}, & x_{33}, & y_1, & y_2, & y_3 & \geq 0
\end{array}
$$

Example: the LP-0_{VCP} model (2/2)

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP\text{-}0_{VCP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{12} = \tilde{x}_{21} = \tilde{x}_{22} = \tilde{x}_{31} = \tilde{x}_{32} = \frac{1}{2},$$

$$\tilde{x}_{13} = \tilde{x}_{23} = \tilde{x}_{33} = 0,$$

$$\tilde{y}_1 = \tilde{y}_2 = 1, \quad \tilde{y}_3 = 0$$

The optimal value $z(\mathrm{LP\text{-}0_{VCP}})$ is 2.

Accordingly we have the lower bound:

$$2 \leq z(\mathrm{ILP_{VCP}})$$

The dual of the LP-0_{VCP} model: the dual variables

Using

- a free dual variable μ_v for each constraint $\sum_{c \in \mathscr{C}} x_{vc} = 1$ $(v \in V)$
- a non-negative dual variable π_{ce} for each constraint $-x_{uc} - x_{vc} + y_c \geq 0$ $(c \in \mathscr{C},\ e = \{u, v\} \in E)$

the dual model of the LP-0_{VCP} model reads as in the next slide.

The $D(\text{LP-0}_{\text{VCP}})$ model

$$\max \quad \sum_{v \in V} \mu_v$$

$$\text{subject to} \quad \mu_v - \sum_{e \in \delta(v)} \pi_{ce} \leq 0, \quad \forall v \in V,\ \forall c \in \mathscr{C},$$

$$\sum_{e \in E} \pi_{ce} \leq 1, \qquad \forall c \in \mathscr{C},$$

$$\mu_v \gtreqless 0, \qquad \forall v \in V,$$

$$\pi_{ce} \geq 0, \quad \forall c \in \mathscr{C},\ \forall e \in E.$$

The obtained LP model is referred to as the $D(\text{LP-0}_{\text{VCP}})$ model.

Example: the $D(\text{LP-0}_{\text{VCP}})$ model (1/3)

The dual $D(\text{LP-0}_{\text{VCP}})$ model for the triangle example is:

$$
\begin{array}{rlllllll}
\max & \mu_1 & + \mu_2 & + \mu_3 & & & & \\
\text{subject to} & \mu_1 & & & -\pi_{112} & -\pi_{113} & & \\
& \mu_1 & & & & & & \\
& \mu_1 & & & & & & \\
& & \mu_2 & & -\pi_{112} & & -\pi_{123} & \\
& & \mu_2 & & & & & \\
& & \mu_2 & & & & & \\
& & & \mu_3 & & -\pi_{113} & -\pi_{123} & \\
& & & \mu_3 & & & & \\
& & & \mu_3 & & & & \\
& & & & \pi_{112} & +\pi_{113} & +\pi_{123} & \\
& & & & & & & \\
& & & & & & & \\
& \mu_1, & \mu_2, & \mu_3 & & & & \\
& & & & \pi_{112}, & \pi_{113}, & \pi_{123}, &
\end{array}
$$

Example: the $D(\text{LP-0}_{\text{VCP}})$ model (2/3)

Continued: the remaining columns and the right-hand sides.

$$
\begin{array}{rcccccccc}
\max & & & & & & & & \\
\text{subject to} & & & & & & & & \\
 & & & & & & & \leq & 0 \\
 & -\pi_{212} & -\pi_{213} & & & & & \leq & 0 \\
 & & & & -\pi_{312} & -\pi_{313} & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & -\pi_{212} & & -\pi_{223} & & & & \leq & 0 \\
 & & & & -\pi_{312} & & -\pi_{323} & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & & -\pi_{213} & -\pi_{223} & & & & \leq & 0 \\
 & & & & & -\pi_{313} & -\pi_{323} & \leq & 0 \\
 & & & & & & & \leq & 1 \\
 & \pi_{212} & +\pi_{213} & +\pi_{223} & & & & \leq & 1 \\
 & & & & \pi_{312} & +\pi_{313} & +\pi_{323} & \leq & 1 \\
 & & & & & & & \geq & 0 \\
 & \pi_{212}, & \pi_{213}, & \pi_{223}, & \pi_{312}, & \pi_{313}, & \pi_{323} & \geq & 0
\end{array}
$$

Example: the $D(\text{LP-0}_{\text{VCP}})$ model (3/3)

The optimal value $z(D(\text{LP-0}_{\text{VCP}}))$ is 2.

An optimal solution $(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\pi}})$ of $D(\text{LP-0}_{\text{VCP}})$ is:

$$\tilde{\mu}_1 = \tilde{\mu}_2 = 1, \quad \tilde{\mu}_3 = 0,$$

$$\tilde{\pi}_{112} = \tilde{\pi}_{212} = \tilde{\pi}_{312} = 1,$$

all the other $\tilde{\pi}_{ce} = 0$

- The primal and the dual optimal values coincide: $z(\text{LP-0}_{\text{VCP}}) = 2 = z(D(\text{LP-0}_{\text{VCP}}))$

On the triangle, both optimal values are 2.

We now show that the same primal and dual solutions are optimal for **every** graph with at least one edge.

Proposition 1: optimal solutions (1/2)

An optimal solution of the primal $\mathrm{LP\text{-}0_{VCP}}$ *model is:*

$$\tilde{x}_{v1} = \tilde{x}_{v2} = \frac{1}{2}, \quad \forall v \in V,$$

$$\tilde{x}_{vc} = 0, \quad \forall v \in V,\ \forall c \in \{3, 4, \ldots, k\},$$

$$\tilde{y}_1 = \tilde{y}_2 = 1,$$

$$\tilde{y}_c = 0, \quad \forall c \in \{3, 4, \ldots, k\},$$

Proposition: optimal solutions (2/2)

and an optimal solution of the dual $D(\text{LP-0}_{\text{VCP}})$ *model is:*

$$\begin{aligned}
\tilde{\mu}_u = \tilde{\mu}_v &= 1, \\
\tilde{\mu}_w &= 0, \quad \forall w \in V \setminus \{u, v\}, \\
\tilde{\pi}_{c\{u,v\}} &= 1, \quad \forall c \in \mathscr{C}, \\
\tilde{\pi}_{ce} &= 0, \quad \forall e \in E \setminus \{\{u, v\}\},
\end{aligned}$$

where $\{u, v\}$ *is any edge in* E*.*

Corollary 1

The optimal value of the primal LP-0_{VCP} *and dual* $D(\text{LP-0}_{\text{VCP}})$ *models is:*

$$z(\text{LP-0}_{\text{VCP}}) = 2 = z(D(\text{LP-0}_{\text{VCP}}))$$

How the proof works

The proof proceeds in three steps:

1. We show that the primal solution of the proposition is a feasible solution for the primal $\mathrm{LP\text{-}0}_{\mathrm{VCP}}$ model;
2. We show that the dual solution of the proposition is a feasible solution for the dual $D(\mathrm{LP\text{-}0}_{\mathrm{VCP}})$ model;
3. We show that both feasible solutions have the same objective function value.

Therefore, by LP Weak Duality, the two solutions are a pair of optimal primal and dual solutions.

Proof (1/4)

1. For every $v \in V$, we have:

$$\sum_{c \in \mathscr{C}} \tilde{x}_{vc} = \frac{1}{2} + \frac{1}{2} = 1$$

Thus the constraints $\sum_{c \in \mathscr{C}} x_{vc} = 1$ are satisfied.

For $c \in \{1, 2\}$ and every $\{u, v\} \in E$, we have:

$$\tilde{y}_c - \tilde{x}_{uc} - \tilde{x}_{vc} = 1 - \frac{1}{2} - \frac{1}{2} = 0$$

and for $c \in \{3, 4, \ldots, k\}$ we have $\tilde{y}_c - \tilde{x}_{uc} - \tilde{x}_{vc} = 0 - 0 - 0 = 0$.

Thus the constraints $-x_{uc} - x_{vc} + y_c \geq 0$ are satisfied.

Proof (2/4)

We have $\tilde{x}_{vc} \geq 0$ and $\tilde{y}_c \geq 0$, for every $v \in V$ and every $c \in \mathscr{C}$, thus the constraints $x_{vc} \geq 0$ and $y_c \geq 0$ are satisfied.

Therefore, the primal solution provides a feasible solution for the primal $\mathrm{LP\text{-}0_{VCP}}$ model.

2. For every $c \in \mathscr{C}$, we have:

$$\sum_{e \in E} \tilde{\pi}_{ce} = 1$$

Thus the constraints $\sum_{e \in E} \pi_{ce} \leq 1$ are satisfied.

Proof (3/4)

For every $c \in \mathscr{C}$ and $w \in \{u, v\}$, we have:

$$\tilde{\mu}_w - \sum_{e \in \delta(w)} \tilde{\pi}_{ce} = 1 - 1 = 0$$

For every $c \in \mathscr{C}$ and $w \in V \setminus \{u, v\}$, we have $\tilde{\mu}_w = 0$ and $\tilde{\pi}_{ce} = 0$ for every $e \in \delta(w)$ (since $\tilde{\pi}_{ce}$ is nonzero only for $e = \{u, v\}$, which is not incident to w), so

$$\tilde{\mu}_w - \sum_{e \in \delta(w)} \tilde{\pi}_{ce} = 0 - 0 = 0.$$

Thus the constraints $\mu_v - \sum_{e \in \delta(v)} \pi_{ce} \leq 0$ are satisfied.

Proof (4/4)

We have $\tilde{\mu}_v \geq 0,\ \forall v \in V$ and $\tilde{\pi}_{ce} \geq 0$ for every $c \in \mathscr{C}$ and every $e \in E$. Thus the constraints $\mu_v \gtreqless 0$ and $\pi_{ce} \geq 0$ are satisfied.

Therefore, the dual solution provides a feasible solution for the dual $D(\text{LP-0}_{\text{VCP}})$ model.

3. For the objective functions we have:

$$\sum_{c \in \mathscr{C}} \tilde{y}_c = 1 + 1 = \sum_{v \in V} \tilde{\mu}_v$$

thus the primal and dual objective function values are equal.

Therefore, by LP Weak Duality, the two solutions are a pair of optimal primal and dual solutions. □

An alternative proof

- This is an alternative proof that does not rely on the dual formulation but instead reasons directly on the minimum objective function value imposed by the constraints

Proposition 2

An optimal solution of the primal $\mathrm{LP\text{-}0_{VCP}}$ *model is:*

$$\begin{aligned} \tilde{x}_{v1} = \tilde{x}_{v2} &= \frac{1}{2}, \quad \forall v \in V, \\ \tilde{x}_{vc} &= 0, \quad \forall v \in V,\ \forall c \in \{3, 4, \ldots, k\}, \\ \tilde{y}_1 = \tilde{y}_2 &= 1, \\ \tilde{y}_c &= 0, \quad \forall c \in \{3, 4, \ldots, k\}, \end{aligned}$$

and the optimal value is $z(\mathrm{LP\text{-}0_{VCP}}) = 2$.

Proof (1/4)

Summing the edge constraints $-x_{uc} - x_{vc} + y_c \geq 0$ over all $c \in \mathscr{C}$, for any edge $\{u, v\} \in E$, gives

$$\sum_{c \in \mathscr{C}} y_c - \sum_{c \in \mathscr{C}} x_{uc} - \sum_{c \in \mathscr{C}} x_{vc} \geq 0$$

$$\implies \sum_{c \in \mathscr{C}} y_c \geq \underbrace{\sum_{c \in \mathscr{C}} x_{uc}}_{=1} + \underbrace{\sum_{c \in \mathscr{C}} x_{vc}}_{=1} = 2,$$

where both sums are equal to 1 by the constraints $\sum_{c \in \mathscr{C}} x_{vc} = 1$.

Proof (2/4)

Hence

$$\sum_{c \in \mathscr{C}} y_c \ \geq \ 2.$$

Therefore every feasible solution of $\mathrm{LP\text{-}0_{VCP}}$ satisfies $\sum_{c \in \mathscr{C}} y_c \geq 2$.

Now consider the point defined in the proposition. For every $v \in V$, we have

$$\sum_{c \in \mathscr{C}} \tilde{x}_{vc} = \tfrac{1}{2} + \tfrac{1}{2} = 1,$$

so the constraints $\sum_{c \in \mathscr{C}} x_{vc} = 1$ hold.

Proof (3/4)

Moreover, for every $\{u, v\} \in E$ and $c \in \{1, 2\}$,

$$\tilde{y}_c - \tilde{x}_{uc} - \tilde{x}_{vc} = 1 - \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

so the constraints $-x_{uc} - x_{vc} + y_c \geq 0$ hold with equality, while for all $c \geq 3$ both sides are zero.

Finally, the constraints $x_{vc} \geq 0$ and $y_c \geq 0$ are satisfied by construction.

Hence the point defined in the proposition is a feasible solution for LP-0_{VCP}.

Proof (4/4)

The objective value of this point is

$$\sum_{c \in \mathscr{C}} \tilde{y}_c = 1 + 1 = 2,$$

which matches the lower bound $\sum_{c \in \mathscr{C}} y_c \geq 2$.

Therefore the point is an optimal solution with objective value $z(\text{LP-0}_{\text{VCP}}) = 2$. □

Summary

The $\mathrm{LP\text{-}0_{VCP}}$ model:

- The binary domains are replaced by $x_{vc} \geq 0$ and $y_c \geq 0$
- Its dual $D(\mathrm{LP\text{-}0_{VCP}})$ has a free variable μ_v for each vertex and a non-negative variable π_{ce} for each color and each edge

The optimal value is 2 for every graph with an edge:

- Primal: $\frac{1}{2}$ on colors 1 and 2 for all the vertices, $y_1 = y_2 = 1$
- Dual: one edge $\{u, v\}$, with $\mu_u = \mu_v = 1$ and $\pi_{c\{u,v\}} = 1$ for every color
- Alternative proof: summing the edge constraints of one edge gives $\sum_{c \in \mathscr{C}} y_c \geq 2$

A second Linear Programming relaxation

Introduction: A second Linear Programming relaxation

- In this section we write a second LP relaxation, the $\mathrm{LP}_{\mathrm{VCP}}$ model, which keeps the upper bounds $y_c \leq 1$
- We solve it and its dual $D(\mathrm{LP}_{\mathrm{VCP}})$ on the triangle
- We prove that, also for this relaxation, the optimal value is 2 for every graph with an edge: adding constraints does not improve the bound

The $\mathrm{LP}_{\mathrm{VCP}}$ model

To obtain a Linear Programming (LP) relaxation of the $\mathrm{ILP}_{\mathrm{VCP}}$ model, we relax the constraints $x_{vc} \in \{0, 1\}$ and $y_c \in \{0, 1\}$ with the following ones:

$$x_{vc} \geq 0, \quad \forall v \in V, \ \forall c \in \mathscr{C},$$

$$y_c \leq 1, \qquad \forall c \in \mathscr{C},$$

$$y_c \geq 0, \qquad \forall c \in \mathscr{C}.$$

This LP model is referred to as the $\mathrm{LP}_{\mathrm{VCP}}$ model.

Example: the $\mathrm{LP}_{\mathrm{VCP}}$ model (1/2)

The $\mathrm{LP}_{\mathrm{VCP}}$ model for the triangle example is:

$$
\begin{array}{rcccccccccccccc}
\text{min} & & & & & & & & & & y_1 & + y_2 & + y_3 & & \\
\text{subject to} & x_{11} & + x_{12} & + x_{13} & & & & & & & & & & = & 1 \\
 & & & & x_{21} & + x_{22} & + x_{23} & & & & & & & = & 1 \\
 & & & & & & & x_{31} & + x_{32} & + x_{33} & & & & = & 1 \\
 & -x_{11} & & & -x_{21} & & & & & & + y_1 & & & \geq & 0 \\
 & -x_{11} & & & & & & -x_{31} & & & + y_1 & & & \geq & 0 \\
 & & & & -x_{21} & & & -x_{31} & & & + y_1 & & & \geq & 0 \\
 & & -x_{12} & & & -x_{22} & & & & & & + y_2 & & \geq & 0 \\
 & & -x_{12} & & & & & & -x_{32} & & & + y_2 & & \geq & 0 \\
 & & & & & -x_{22} & & & -x_{32} & & & + y_2 & & \geq & 0 \\
 & & & -x_{13} & & & -x_{23} & & & & & & + y_3 & \geq & 0 \\
 & & & -x_{13} & & & & & & -x_{33} & & & + y_3 & \geq & 0 \\
 & & & & & & -x_{23} & & & -x_{33} & & & + y_3 & \geq & 0 \\
 & & & & & & & & & & y_1 & & & \leq & 1 \\
 & & & & & & & & & & & y_2 & & \leq & 1 \\
 & & & & & & & & & & & & y_3 & \leq & 1 \\
 & x_{11}, & x_{12}, & x_{13}, & x_{21}, & x_{22}, & x_{23}, & x_{31}, & x_{32}, & x_{33}, & y_1, & y_2, & y_3 & \geq & 0
\end{array}
$$

Example: the $\mathrm{LP}_{\mathrm{VCP}}$ model (2/2)

An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_{\mathrm{VCP}})$ is:

$$\tilde{x}_{11} = \tilde{x}_{12} = \tilde{x}_{21} = \tilde{x}_{22} = \tilde{x}_{31} = \tilde{x}_{32} = \frac{1}{2},$$

$$\tilde{x}_{13} = \tilde{x}_{23} = \tilde{x}_{33} = 0,$$

$$\tilde{y}_1 = \tilde{y}_2 = 1, \quad \tilde{y}_3 = 0$$

The optimal value $z(\mathrm{LP}_{\mathrm{VCP}})$ is 2. Accordingly we have the same lower bound:

$$2 \leq z(\mathrm{ILP}_{\mathrm{VCP}})$$

The dual of the $\mathrm{LP}_{\mathrm{VCP}}$ model: the dual variables

Using

- a free dual variable μ_v for each constraint $\sum_{c \in \mathscr{C}} x_{vc} = 1$ $(v \in V)$
- a non-negative dual variable π_{ce} for each constraint $-x_{uc} - x_{vc} + y_c \geq 0$ $(c \in \mathscr{C},\ e = \{u, v\} \in E)$
- a non-positive dual variable λ_c for each constraint $y_c \leq 1$ $(\forall c \in \mathscr{C})$

the dual model of the $\mathrm{LP}_{\mathrm{VCP}}$ model reads as in the next slide.

The $D(\mathrm{LP}_{\mathrm{VCP}})$ model

$$\max \quad \sum_{v \in V} \mu_v + \sum_{c \in \mathscr{C}} \lambda_c$$

$$\begin{aligned}
\text{subject to} \quad & \mu_v - \sum_{e \in \delta(v)} \pi_{ce} \leq 0, && \forall v \in V,\ \forall c \in \mathscr{C}, \\
& \sum_{e \in E} \pi_{ce} + \lambda_c \leq 1, && \forall c \in \mathscr{C}, \\
& \mu_v \gtreqless 0, && \forall v \in V, \\
& \pi_{ce} \geq 0, && \forall c \in \mathscr{C},\ \forall e \in E, \\
& \lambda_c \leq 0, && \forall c \in \mathscr{C}.
\end{aligned}$$

The obtained LP model is referred to as the $D(\mathrm{LP}_{\mathrm{VCP}})$ model.

Example: the $D(\mathrm{LP}_{\mathrm{VCP}})$ model (1/3)

The dual $D(\mathrm{LP}_{\mathrm{VCP}})$ model for the triangle example is:

$$
\begin{array}{rlllllllllll}
\max & \mu_1 & +\mu_2 & +\mu_3 & & & & & & & & \\
\text{subject to} & \mu_1 & & & -\pi_{112} & -\pi_{113} & & & & & & \\
& \mu_1 & & & & & & -\pi_{212} & -\pi_{213} & & & \\
& \mu_1 & & & & & & & & & & \\
& & \mu_2 & & -\pi_{112} & & -\pi_{123} & & & & & \\
& & \mu_2 & & & & & -\pi_{212} & & -\pi_{223} & & \\
& & \mu_2 & & & & & & & & & \\
& & & \mu_3 & & -\pi_{113} & -\pi_{123} & & & & & \\
& & & \mu_3 & & & & & -\pi_{213} & -\pi_{223} & & \\
& & & \mu_3 & & & & & & & & \\
& & & & \pi_{112} & +\pi_{113} & +\pi_{123} & & & & & \\
& & & & & & & \pi_{212} & +\pi_{213} & +\pi_{223} & & \\
& & & & & & & & & & & \\
& \mu_1, & \mu_2, & \mu_3 & & & & & & & & \\
& & & & \pi_{112}, & \pi_{113}, & \pi_{123}, & \pi_{212}, & \pi_{213}, & \pi_{223}, & &
\end{array}
$$

Example: the $D(\mathrm{LP}_{\mathrm{VCP}})$ model (2/3)

Continued: the remaining columns and the right-hand sides.

$$
\begin{array}{rcccccccc}
\max & & & & +\lambda_1 & +\lambda_2 & +\lambda_3 & & \\
\text{subject to} & & & & & & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & -\pi_{312} & -\pi_{313} & & & & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & -\pi_{312} & & -\pi_{323} & & & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & & & & & & & \leq & 0 \\
 & & -\pi_{313} & -\pi_{323} & & & & \leq & 0 \\
 & & & & +\lambda_1 & & & \leq & 1 \\
 & & & & & +\lambda_2 & & \leq & 1 \\
 & \pi_{312} & +\pi_{313} & +\pi_{323} & & & +\lambda_3 & \leq & 1 \\
 & & & & & & & \geq & 0 \\
 & \pi_{312}, & \pi_{313}, & \pi_{323} & & & & \geq & 0 \\
 & & & & \lambda_1, & \lambda_2, & \lambda_3 & \leq & 0
\end{array}
$$

Example: the $D(\mathrm{LP}_{\mathrm{VCP}})$ model (3/3)

The optimal value $z(D(\mathrm{LP}_{\mathrm{VCP}}))$ is 2.

An optimal solution $(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\pi}}, \tilde{\boldsymbol{\lambda}})$ of $D(\mathrm{LP}_{\mathrm{VCP}})$ is:

$$\tilde{\mu}_1 = \tilde{\mu}_2 = 1, \quad \tilde{\mu}_3 = 0,$$

$$\tilde{\pi}_{112} = \tilde{\pi}_{212} = \tilde{\pi}_{312} = 1,$$

all the other $\tilde{\pi}_{ce} = 0$,

$$\tilde{\lambda}_1 = 0, \quad \tilde{\lambda}_2 = 0, \quad \tilde{\lambda}_3 = 0$$

Valid constraints that are not needed

- The constraints:

$$x_{vc} \leq 1, \qquad \forall v \in V,\ \forall c \in \mathscr{C},$$

  are also valid since, for every $v \in V$ and every $c \in \mathscr{C}$, we have:

$$x_{vc} \in \{0,1\} \qquad \Rightarrow \qquad x_{vc} \leq 1$$

- But they are implied by the constraints $\sum_{c \in \mathscr{C}} x_{vc} = 1$

Proposition 3: optimal solutions (1/2)

An optimal solution of the primal $\mathrm{LP}_{\mathrm{VCP}}$ *model is:*

$$\tilde{x}_{v1} = \tilde{x}_{v2} = \frac{1}{2}, \quad \forall v \in V,$$

$$\tilde{x}_{vc} = 0, \quad \forall v \in V,\ \forall c \in \{3, 4, \ldots, k\},$$

$$\tilde{y}_1 = \tilde{y}_2 = 1,$$

$$\tilde{y}_c = 0, \quad \forall c \in \{3, 4, \ldots, k\},$$

Proposition: optimal solutions (2/2)

and an optimal solution of the dual $D(\mathrm{LP}_{\mathrm{VCP}})$ *model is:*

$$\begin{aligned}
\tilde{\mu}_u = \tilde{\mu}_v &= 1, \\
\tilde{\mu}_w &= 0, \quad \forall w \in V \setminus \{u, v\}, \\
\tilde{\pi}_{c\{u,v\}} &= 1, \quad \forall c \in \mathscr{C}, \\
\tilde{\pi}_{ce} &= 0, \quad \forall e \in E \setminus \{\{u, v\}\}, \\
\tilde{\lambda}_c &= 0, \quad \forall c \in \mathscr{C},
\end{aligned}$$

where $\{u, v\}$ *is any edge in* E*.*

Proof

The proof is similar to the previous one.

In addition it is necessary to observe that:

$$\tilde{y}_c \leq 1, \quad \forall c \in \mathscr{C}.$$

Accordingly, also the constraints $y_c \leq 1$ are satisfied. □

Corollary 2

The optimal value of the primal $\mathrm{LP}_{\mathrm{VCP}}$ *and dual* $D(\mathrm{LP}_{\mathrm{VCP}})$ *models is:*

$$z(\mathrm{LP}_{\mathrm{VCP}}) = 2 = z(D(\mathrm{LP}_{\mathrm{VCP}}))$$

Summary

The $\mathrm{LP}_{\mathrm{VCP}}$ model:

- The binary domains are replaced by $x_{vc} \geq 0$, $y_c \leq 1$ and $y_c \geq 0$
- Its dual $D(\mathrm{LP}_{\mathrm{VCP}})$ has, in addition, a non-positive variable λ_c for each color
- The constraints $x_{vc} \leq 1$ are valid, but implied by $\sum_{c \in \mathscr{C}} x_{vc} = 1$

The same bound:

- The same primal solution, and the same dual solution with $\lambda_c = 0$, are optimal
- $z(\mathrm{LP}_{\mathrm{VCP}}) = 2 = z(D(\mathrm{LP}_{\mathrm{VCP}}))$ for every graph with an edge

Typical mistakes

Introduction: Typical mistakes

- In this section we list the most common mistakes on the $\mathrm{ILP}_{\mathrm{VCP}}$ model and its relaxations
- They concern the choice of k, the LP bound, the activation variables, the symmetry and the dual solution
- Recognizing them helps to write and to use the model correctly

Observation 1: typical mistakes (1/2)

- ***Choosing*** *k* ***too small.*** *The number k of colors must be at least the chromatic number of the graph: with fewer colors the model has no feasible solution. The value $k = n$ is always valid.*
- ***Trusting the LP bound.*** *The optimal value of both relaxations is 2 for every graph with at least one edge, even for the complete graph on n vertices, whose optimal value is n.*

Observation: typical mistakes (2/2)

- ***Believing that the model counts only the colors that are really used.*** *The constraints allow $y_c = 1$ for a color that no vertex takes: it is the objective function that switches such colors off.*
- ***Ignoring the symmetry.*** *Permuting the colors of an optimal solution gives another optimal solution: the $\mathrm{ILP}_{\mathrm{VCP}}$ model has $k!$ equivalent copies of every coloring.*
- ***Writing the dual solution with a wrong edge.*** *The dual values $\tilde{\pi}_{ce}$ are non-zero only for one edge $e = \{u, v\}$, and for every color c: all the other edges have value zero.*

Summary

When writing the model:

- Choose k at least equal to the chromatic number: $k = n$ is always valid
- The objective function, not the constraints, switches off the unused colors
- Permuting the colors gives equivalent copies of the same coloring

When using the relaxations:

- The LP bound is 2 for every graph with an edge, even for the complete graph
- In the dual solution, only one edge has non-zero values $\tilde{\pi}_{ce}$

In summary

Introduction: In summary

- We collect the key ideas of the chapter in a few points
- We close with the dashboard of the triangle example: the bounds on $z(\mathrm{ILP}_{\mathrm{VCP}})$ computed in this chapter

Summary (1/2)

- The $\mathrm{ILP}_{\mathrm{VCP}}$ model has $k \cdot n$ variables x_{vc} (vertex v takes color c) and k variables y_c (color c is used); the edge constraints forbid equal colors on adjacent vertices
- The two LP relaxations have the same optimal value 2 for every graph with an edge: put $\frac{1}{2}$ on two colors for all the vertices
- The dual solution selects one edge $\{u, v\}$ and gives value 1 to μ_u, μ_v and $\pi_{c\{u,v\}}$ for every color
- The bound 2 is very weak: a formulation that must color graphs needs additional valid inequalities

Summary (2/2)

Dashboard of the triangle example. Bounds on $z(\mathrm{ILP}_{\mathrm{VCP}})$ computed in this chapter:

Bound	Value	From
First relaxation	2	$z(\mathrm{LP\text{-}0}_{\mathrm{VCP}})$
Second relaxation	2	$z(\mathrm{LP}_{\mathrm{VCP}})$
Optimal value	3	color classes $\{1\}$, $\{2\}$, $\{3\}$

The gap between 2 and 3 is small here, but it grows with the size of the complete graph: 2 against n.